Reactive-film runtime probe at NCCC Position 1

Numerical reachability and manuscript-adoption decision

MEA Absorption Column project

2026-09-01

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1 Question

Does the derivative-enabled reactive-film formulation reach a converged numerical solution at retained NCCC Position 1, and are its values ready for manuscript-level physical claims?

2 Retained calculation

Quantity	Retained value
Position	1
Temperature	318.150 K
Pressure	109,500 Pa
CO2 loading	0.249996 mol CO2 mol-1 MEA
Analytical MEA concentration	4.889310 mol L-1
Initial mesh	21 points
Solver result	Converged to desired accuracy
Solver iterations	6
Final mesh	355 points
Wall time	592.179 s
ODE Jacobian	Exact ePC-SAFT activity chain rule
Scientific admission	No

The calculation used the `MEA_CO2_H2O_retained_predictive` method-development dataset and retained Position 1 speciation. It did not use an accepted predictive parameter packet. The finite-rate coefficients and diffusivities are provisional because the available source candidates failed the scientific-admission checks listed below.

3 Numerical results

Quantity	Value
Bulk liquid CO2 fugacity	35.757415 Pa
Interface liquid CO2 fugacity	3,075.239074 Pa
Bulk vapor CO2 fugacity	3,077.376293 Pa
Vapor-to-interface fugacity gap	2.137220 Pa
Interface CO2 flux	5.312554e-5 mol m-2 s-1
Predicted axial CO2 flux	0.003914766 mol s-1 m-1 packed height
Retained column flux comparator	0.451980846 mol s-1 m-1 packed height
Comparator / probe flux	115.4554
Probe flux as fraction of comparator	0.8661%
Interface / bulk liquid fugacity	86.0028

The interface liquid and bulk vapor fugacities differ by 0.06945% of the bulk vapor fugacity. This small interfacial driving force accompanies a large liquid-film fugacity rise from the bulk state. The retained column value is a diagnostic comparator from a different formulation, not an independent measurement and not a thermodynamic-parameter fitting target.

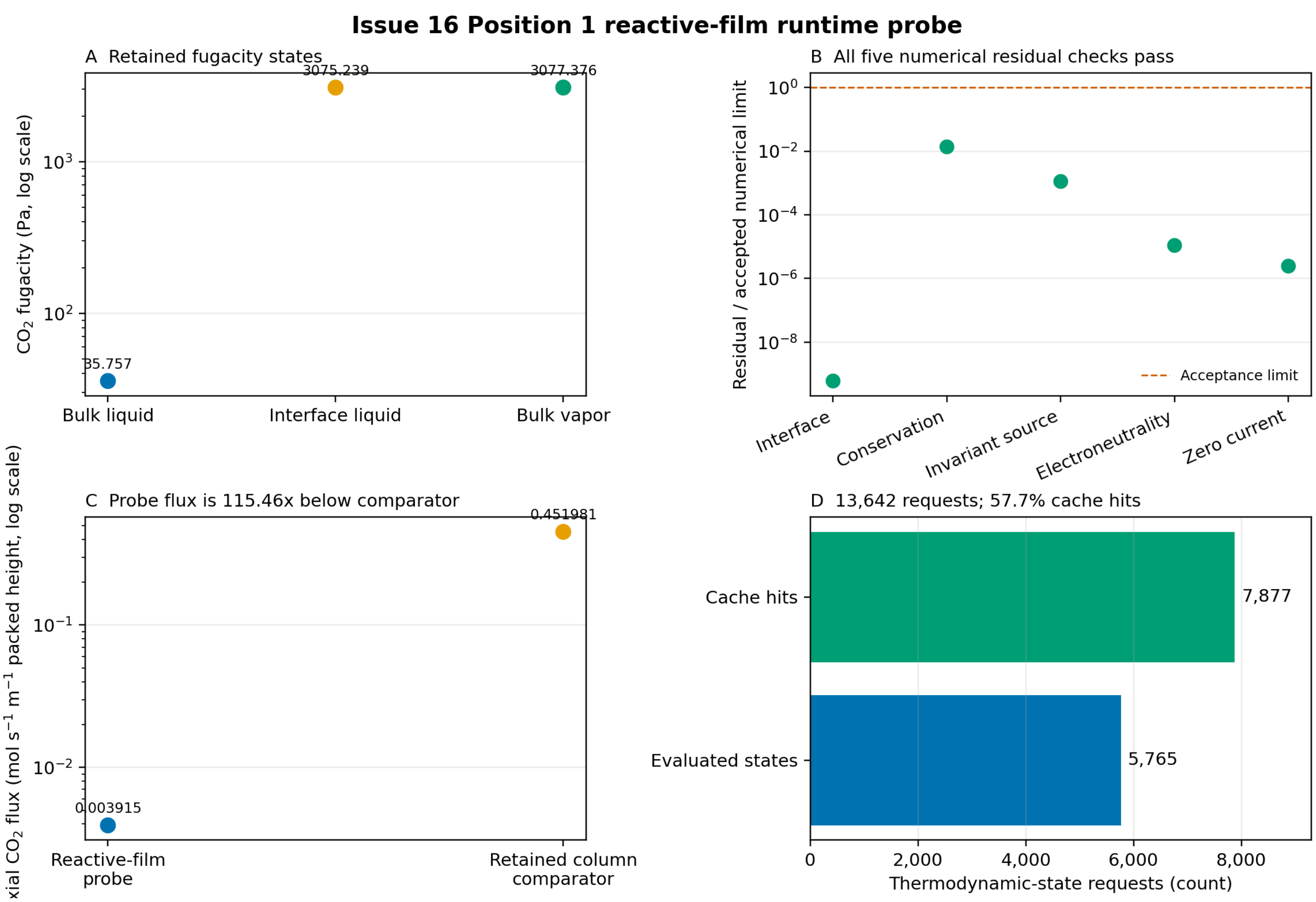

Figure 1: The four panels report the retained fugacity states, residual-to-limit ratios, axial flux comparison, and exact thermodynamic-state work accounting. Each axis is categorical because only one retained physical state was evaluated.

3.1 Numerical residual certificate

Check	Maximum residual	Accepted limit	Residual / limit
Interface closure	5.960091e-17	1.0e-7	5.960091e-10
Conservation	1.364461e-9	1.0e-7	1.364461e-2
Invariant source	1.114721e-15	1.0e-12	1.114721e-3
Electroneutrality	1.087048e-17	1.0e-12	1.087048e-5
Zero current	2.435875e-18	1.0e-12	2.435875e-6

All five retained numerical residuals are below their accepted limits. The largest normalized residual is conservation at 0.01364 of its limit. This establishes numerical convergence for this declared attempt; it does not establish scientific validity of the provisional rate and transport inputs.

3.2 Thermodynamic work accounting

Counter	Count
Thermodynamic-state requests	13,642
Derivative-rich state evaluations	5,765
Exact-composition cache hits	7,877
Cache-hit fraction	57.74%
Density-anchor misses	1
Density-anchor hits	5,764

The request accounting closes exactly: 5,765 evaluations plus 7,877 cache hits equal 13,642 requests. On this machine and execution path, the retained run completed in 592.179 s; the earlier 300 s watchdog interrupted this calculation before completion and was not evidence of solver failure. This single observation does not prescribe a general timeout.

4 Scientific limitations

Limitation	Retained status	Consequence
MEA concentration basis	4.889310 mol L-1 does not equal the discrete 5 mol L-1 source label	Source-domain mapping is unresolved
F1/F2 finite-rate coefficients	Rejected: source unit inconsistency	Rates are not scientifically admitted
F3 finite-rate coefficient	Rejected: primary coefficient source unavailable	Rates are not scientifically admitted
Diffusivities	All numeric candidates rejected	Transport is not scientifically admitted
Parameter packet	Method-development dataset only	No predictive thermodynamic claim
Physical-state coverage	One retained Position 1 state	No position, loading, or temperature generalization
Repetition	One completed run	No runtime or repeatability distribution
Column validation	Not performed with this formulation	No capture-performance claim

5 Conclusion and manuscript-adoption decision

The result is a successful numerical-reachability demonstration. At one retained Position 1 state, the derivative-enabled reactive-film calculation converged in 6 iterations on 355 final mesh points, met every retained residual limit, and completed in 592.179 s on this machine and execution path. The public density anchor and exact activity-chain-rule derivatives therefore support a working nonlinear film solve rather than a pre-solver rejection or an established solver failure.

The numerical values should **not** be adopted as manuscript-level physical or predictive results. The row is explicitly scientifically inadmissible, uses unresolved kinetics and diffusivities, covers only one state, has no repeatability or multi-position evidence, and has not been integrated into a validated column campaign. In particular, the 115.4554-fold gap from the retained column comparator must not be used to retune ePC-SAFT parameters or presented as model accuracy; the comparator is not an independent experiment and the film's source terms and transport basis are not yet admitted.

The only defensible near-term use is internal method-development evidence: the intended nonlinear reactive-film formulation was numerically reachable in this retained run. Future campaign timeouts must be chosen from their own runtime evidence rather than this single observation. Manuscript promotion should wait until admitted kinetics, diffusivities, and concentration-basis mapping are supplied; the same formulation passes repeated runs and multiple declared positions; and column-level outputs are compared with independent rate or capture evidence.

An independent CSE Review on 2026-09-01 checked this working-tree analysis against repository base commit 700f6212 fb20ab4b2b9045c1b9da9d23a5acb88a. It passed for internal numerical-reachability use after provenance, units, and runtime wording were corrected. The review does not admit the provisional scientific inputs or promote this notebook into the manuscript; investigator promotion remains required after the stated scientific gates pass.

6 Provenance

The QMD is hand-authored from the retained Position 1 CSV, summary, and identity manifest. The figure script reads only the retained CSV. Rendering does not run the film, column, chemistry, or ePC-SAFT calculation.

- Probe CSV SHA-256: 0e070f1731faefcecbbe683b5b583203f3829d299937650e1aa1381882637be0
- Probe summary SHA-256: 0f93215d233692ecb6d369eb425fbd8c88bacf0c0d9e0afa96d2ccc2fa4eb601
- Identity manifest: `analyses/nccc_validation/inputs/issue16_provisional_reactive_film_identity.json`
- Identity-manifest SHA-256: 62779445ba2ba37debb12ba8b27af8711bb05c58e52f0848c71f9f1351c548a0
- ePC-SAFT Engine commit: 9d7bebe5760bb425a08a9211d5dfd353e8693832
- Wheel SHA-256: 7c4be78427cd9d5f70fd60dc63f40d93b7e37945ae7a124dd8d698d94190c9f7
- Engine core SHA-256: edbac80c39a17db588adccedd4c45e578daccba583bf18f0917d2bc00911b899
- Parameter-document SHA-256: 66837ab1c9625f28cf143a8adfea5855db70d3e070d44258d7e6f5f3e133c63b
- Reactive-speciation table SHA-256: 36ae5bf625f076e5f31f8a3f26cccd40f9d3d33526190fac97e46e887507bdb4